

2023



Migrating Critical ERP Workloads to AWS Using SAP on AWS with 3M Company

Learn how global manufacturer 3M Company migrated critical ERP workloads at large scale to SAP on AWS.

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Cut over

hundreds of terabytes of data in under 20 hours

Migrated

approximately 1,200 servers in 15 months

Improved

scalability and increased resiliency

Saved

more than 50% on SAP S/4HANA license costs

Reduced

infrastructure management tasks for e



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[3M Company](#) (3M) needed to migrate its enterprise resource planning (ERP) solution from aging data centers to the cloud. The global manufacturing and material science company housed its ERP and other applications in data centers that couldn't support the levels of scalability, stability, and sustainability that 3M required. To find better, more scalable options for hosting its critical workloads, 3M looked to the cloud.

In 2020, 3M began migrating at scale to Amazon Web Services (AWS). Its ERP was migrated to [SAP on AWS](#), which offers proven approaches backed by expert experience supporting SAP customers in the cloud on AWS. Using AWS, 3M completed the migration of its ERP solution—approximately 1,200 servers—in only 15 months, 3 months ahead of schedule. Now, the company has improved resiliency and is optimizing costs in the cloud.



Opportunity | Working alongside AWS Professional Services and AWS Partners to Migrate 3M at Scale

A diversified, global manufacturing and material science company, 3M produces a variety of products, from adhesives and building materials to cleaning and medical supplies. During the COVID-19 pandemic, 3M ramped up production of masks to help meet global demand. To manage its operations, 3M had a large ERP environment running across approximately 1,200 servers. These workloads were hosted in aging data centers, which could not provide the scalability, resilience, or stability that 3M needed for its ERP solution.

3M Enterprise IT began to weigh its options and selected AWS as its preferred cloud services provider. “3M had experience using AWS in one of our business units,” says Kyle Hammer, director of cloud trans
“With our previous experience and the proven results that we had using AWS, plus the design methodology that AWS presented, we knew that using AWS was the solution we needed to move forward with our



SAP ERP migration.” To facilitate the massive migration, 3M collaborated with [AWS Professional Services](#), a global team of experts that can help customers realize desired business outcomes using AWS. Engineers from 3M, AWS Professional Services, and AWS Partners worked together to thoughtfully plan and test the migration for 8 months before the process began.

Solution | Improving Resilience and Saving Millions of Dollars Using SAP on AWS

In 2020, 3M began its migration to AWS, prioritizing its ERP environments. In total, 3M migrated 11,000 servers, with approximately 1,200 servers for its SAP environment. Instead of slowly building up to the SAP workloads—the largest single portion of 3M’s total migration—3M set the goal of migrating its ERP in the first 18 months of the migration process. The engineering team at 3M built out the design, the virtual data centers, the regions, and the Availability Zones where the ERP environment would be deployed. Then, 3M piloted several iterations of the migration process for the SAP workloads, tested the performance and timing of the migration, and evaluated the results together with the experts at AWS Professional Services. Then, 3M worked with the AWS team to adjust the process to find the most efficient way to migrate workloads with the least amount of business disruption. In the final cutover, the ERP systems—which consisted of hundreds of terabytes of data—were migrated in less than 20 hours. This cutover took place during the COVID-19 pandemic while 3M was manufacturing millions of respirators and masks. “We were able to migrate the workloads that were part of our mask-manufacturing system with minimal disruption,” says Hammer. The SAP migration was completed in only 15 months.

Using SAP on AWS, 3M has greater flexibility and scalability to respond to changes in demand. 3M can scale its compute capacity up or down as needed using [Amazon Elastic Compute Cloud](#) (Amazon EC2), which offers secure and resizable compute capacity for virtually any workload. This flexible scalability means that the company can react to changes in demand in near real time. “We can adjust very quickly to new or different workload demands to solve business challenges or to align with specific releases from SAP, where we might need to scale up for significant data entry or processing,” says Hammer. “On premises, that would have taken many weeks and lots of planning, plus lots of time from our operations teams to build and deploy and reinstall servers.” Now, on AWS, 3M also takes advantage of scheduling and automation to optimize compute workloads, especially during peak periods, and eliminates unnecessary resources to optimize costs. 3M has already saved millions of dollars by rightsizing compute instances, reducing storage, scaling down unused resources, and choosing optimal solutions for storage and back



3M increased the resiliency of its systems using SAP on AWS. In collaboration with the AWS team, 3M introduced additional layers of resiliency into the system design, including hosting the ERP in multiple Availability Zones on AWS and introducing new recovery capabilities. “We can now operate with much more resiliency than we had previously,” says Hammer. “On AWS, we’ve gained flexibility and scale, and we have the availability and performance that we have needed for our SAP environment.” In the cloud, 3M reduced its technical debt and its dependency on hardware. Now, 3M can mature the solutions that support the SAP environment more quickly. “That’s all because of the technology and capability that we have in the cloud,” says Hammer.

Using AWS reduces the burden on employees to manage the infrastructure that supports 3M environments. Now, employees can focus on other tasks. “We can focus energy on new opportunities, resiliency, performance, and improving the application instead of troubleshooting hardware and monitoring the performance of the platforms that were supporting the SAP environment on premises,” says Hammer.

Outcome | Modernizing Infrastructure and Applications in the Cloud

3M completed its migration at scale in November 2022, and the company is now working to modernize its applications using cloud-native services on AWS. “The end result of this migration to AWS is that we migrated quickly, enhanced our landscape, increased our resiliency, and gained the scalability that we needed from an application workload perspective,” says Hammer.

About 3M Company

3M Company is a manufacturing company that uses science to improve lives and solve the world’s toughest challenges. 3M has corporate operations in 70 countries and sales in over 200.

AWS Services Used

SAP on AWS

Get more flexibility and value out of your SAP investments with the world’s most secure, reliable, and extensive cloud infrastructure, 200+ AWS services to innovate, and, purpose-built SAP automation tooling to reduce risk and simplify operations.

[Learn more »](#)

AWS Professional Services



AWS Professional Services offerings help you achieve specific outcomes related to enterprise cloud adoption. Each offering delivers a set of activities, best practices, and documentation reflecting our experience supporting hundreds of customers in their journey to the AWS Cloud.

[Learn more »](#)

Amazon EC2

Amazon Elastic Compute Cloud (Amazon EC2) offers the broadest and deepest compute platform, with over 750 instances and choice of the latest processor, storage, networking, operating system, and purchase model to help you best match the needs of your workload.

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